MIL-E-22200/6C
15 March 1976
SUPERSEDING
MIL-E-22200/6B
28 March 1975
(See 6.4)

MILITARY SPECIFICATION

ELECTRODES, WELDING, MINERAL COVERED, LOW-HYDROGEN, MEDIUM AND HIGH TENSILE STEEL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers low-hydrogen types of covered electrodes for use in welding of low and medium carbon and high tensile steels. Weld metal mechanical properties, including notch toughness, are not adversely affected by thermal stress relieving.

1.2 Classification. Electrodes shall be furnished in the types, classes, and sizes shown in table I, as specified (see 6.2).

Table I - Classification.

Type[1/2/]	Class	Position of welding	Sizes
MIL-7015 MIL-7016	1	All	3/32, 1/8, 5/32, 3/16
MIL-8015-C3 MIL-8016-C3	2	Horizontal fillets and flats	7/32, 1/4
MIL-10015 MIL-10016	3	Flat	5/16

[1/] Type classifications of electrodes will be determined according to the capability of the electrode in meeting the tests specified herein, when used in the positions of welding specified in table I and section 4. An electrode shall be approved under ONE type classification; but an electrode shall NOT be approved if it differs in type from that stated in the manufacturer's request for test. The position description indicates the maximum size qualified under this specification. However, reference should be made to the Qualified Products List (QPL) for the selection of electrodes for all positions of welding.

[2/] The suffix RC added to the MIL-type pertains to restrictions of the copper content (see 3.4). All requirements, tests, and acceptance criteria remain otherwise unchanged.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY
MIL-W-10430 - Welding Rods and Electrodes; Preparation for Delivery of.
MIL-E-22200 - Electrodes, Welding, Covered; General Specification for.
MIL-S-22698 - Steel Plate, Carbon, Structural, for Ships.

STANDARDS

MILITARY
MIL-STD-271 - Nondestructive Testing Requirements for Metals.
MIL-STD-418 - Mechanical Tests for Welded Joints.

FSC 3439

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:23Z
Check the source to verify that this is the current version before use.

PUBLICATION

MILITARY
NAVSHIPS 0900-LP-003-9000 - Radiographic Standards for Production and Repair Welds.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION INCORPORATED, AGENT
National Motor Freight Classification Rules.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., 1616 P Street, N.W., Washington, DC 20036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A131 - Structural Steel for Ships.
A370 - Methods and Definitions for Mechanical Testing of Steel Products.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)
Z49.1 - Safety in Welding and Cutting.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Electrodes furnished under this specification shall conform to the requirements of MIL-E-22200 except as specified herein.

3.2 Covering composition. The chemical composition of the coverings is optional with the manufacturer. The water content of the coverings shall not exceed the percent-by-weight values specified in table II.

Table II - Water content.

Type	Percent by weight maximum
1/ MIL-7015, 7016 MIL-6015-C3 MIL-8015-C3	0.40
MIL-10015, 10016	0.20

1/ Maximum water content not applicable when ordered in class 5 container.

3.3 Groove welds.

3.3.1 Mechanical properties. The mechanical properties of deposited weld metal shall be as specified in table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:23Z
Check the source to verify that this is the current version before use.

Table III - Mechancial properties.

Type	Ultimate tensile strength (as-welded)	Yield strength[1/]		Elongated (min.) as-welded and stress-relieved	Charpy V-notch impact properties[2/3/] (as-welded and stress-relieved)	
		As-welded	Stress-relieved		Energy (Ft/lbs)	Temperature °F
	P.s.i. (Min.)	P.s.i. (Min.)	P.s.i. (Min.)	Percent in 2 inches	Minimum average	Minus
MIL-7015 MIL-7016	70,000	60,000	55,000	24	20	20
MIL-8015-C3 MIL-8016-C3	80,000	70,000	65,000	24	20	40
MIL-10015 MIL-10016	100,000	90,000	80,000	20	20	0

1/ The yield strength shall be determined by the offset method (offset = 0.2 percent).

2/ Impact properties shall be determined by Charpy-V specimens made from material stress-relieved in accordance with note 8 to figure 1.

3/ Stress-relieved and as-welded impact properties shall be determined for qualification of the electrodes for the QPL.

3.4 Chemical composition of deposited weld metal. Chemical composition of deposited weld metal shall be as specified in table IV.

Table IV - Chemical composition (percent)[1/] of deposited weld metal.[2/]

	MIL-7015 and MIL-7016	MIL-8015-C3 and MIL-8016-C3	MIL-10015 and MIL-10016
Carbon	[3/] 0.12	[3/] 0.12	[3/] 0.10
Manganese	.40-1.10	.40-1.10	.70-1.10
Silicon	.20-.60	.20-.60	.05-.60
Phosphorus	.025	.025	.025
Sulfur	.03	.03	.030
Chromium	.15	.15	.15
Nickel	[3/] .25	[3/] .80-1.10	1.40-1.80
Molybdenum	[3/] .35	[3/] .35	.35
Vanadium	.05	.05	.20

1/ Single values shown are maximum percentages, except where otherwise specified.

2/ Use of the suffix RC, e.g., MIL-8015-C3 RC, indicates that the maximum copper content of the deposited weld metal shall be 0.15 percent. The remaining basic compositional requirements remain unchanged for the specific MIL-type (see 6.2).

3/ Optional - An increase of 0.1 percent molybdenum for each 0.01 percent carbon below 0.08 percent is acceptable.

3.5 Nominal lengths. The specified nominal lengths of electrodes shall be in accordance with table V.

Table V - Electrode lengths.

Sizes	Electrode lengths (end-grip)
Inch	Inches
3/32	9, 12 or 14 (see 6.2)
1/8	9 or 14 (see 6.2)
5/32, 3/16	14
7/32, 1/4, 5/16	18

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:23Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 The quality assurance provisions shall be in accordance with MIL-E-22200 except as specified herein.

4.2 Qualification tests. Qualification tests shall be as specified in 4.2 of MIL-E-22200 and tables IX and X herein.

4.3 Comparison inspection. In addition to the examination and tests required in 4.3 of MIL-E-22200, tests specified in table VI shall be conducted as specified in 4.3 of MIL-E-22200, when required by the Naval Ship Engineering Center (NAVSEC).

Table VI - Comparison inspection.

Types	Sizes	Tests	Requirements
All All	All All[1/]	4.6.8 (MIL-E-22200) Weld test No. 1 (see table IX herein)	3.4 and table IV herein 3.4.1 and table IV herein

[1/] Impact tests not required.

4.4 Quality conformance inspection.

4.4.1 Lot. A lot of electrodes shall be as specified in 4.4.1 and 4.4.6 of MIL-E-22200 for inspection level C.

4.4.2 Sampling for quality conformance inspection. Sampling for quality conformance inspection shall be in accordance with 4.4.7 through 4.4.8.2 of MIL-E-22200 for tests specified in MIL-E-22200 where required and as specified in 4.4.2 for quality conformance weld tests.

4.4.3 Quality conformance inspection. Samples shall be selected as specified in 4.4.2. One sample representing one wet mix of covering mixture in each lot shall be subjected to all the tests required by table VII (i.e., X-ray, mechanical, and complete chemical analysis). Samples representing each of the remaining wet mixes of covering mixture in a lot shall be tested as specified in table VIII, to assure equivalency as required by 4.4.4.2 of MIL-E-22200.

Table VII - Quality conformance inspection.

Types	Tests	Procedures
All	Test No. 3 and 8[1/ 2/] Table X	Table IX Table X
All with suffix RC	Test No. 3 and 8[1/ 2/] Table X, plus a copper analysis	Tables IX and X

[1/] Impact properties shall be determined only when required by the procurement document (see 6.2).

[2/] Stress-relieved mechanical and impact properties shall be determined only when required by the procurement document (see 6.2).

Table VIII - Wet mix equivalency tests for quality conformance inspection.

Types	Tests	Procedures
All	Assure equivalency[1/]	4.4.4.2 of MIL-E-22200 Manufacturer's discretion

[1/] See 4.4.4.2 of MIL-E-22200.

4.4.4 Unsatisfactory test results. If the results of the first test are unsatisfactory, two retests shall be permitted. The results of both retests shall be satisfactory for quality conformance inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:23Z
Check the source to verify that this is the current version before use.

4.5 Test procedures.

4.5.1 Weld tests. Table IX summarizes the weld tests required under this specification for qualification testing of electrodes and for the inspection specified in tables VI, VII and VIII. Medium steel base material in accordance with MIL-S-22698 or ASTM A131 may be used for all electrode tests.

Table IX - Summary of required weld tests.[1/]

Weld test No.	Type	Diameter sizes							Weld joint design procedure	Welding position[2/]	Tests			Chemical	Requirements
		3/32	1/8	5/32	3/16	7/32	1/4	5/16			X-ray	Tensile[4/]	Impact		
3	All	X	X	X	X	X	X	X	Figure 1 and notes	F	3/	X	X	Chemical analysis[5/]	3.7.1 of MIL-E-22200 and tables III and IV herein
8a	All	X ---	X ---	X ---	X ---	--- X	--- X	--- X	Figure 2 and notes	V F	X X[3/]	--- ---	--- ---	--- ---	3.7.1 of MIL-E-22200

1/ For qualification testing of the electrodes for the QPL, all electrode types shall be tested using direct current (d.c.) reverse polarity (r.p.). In addition, MIL-7016 and MIL-9016-C3 electrode types shall be tested using alternating current (a.c.). For comparison inspection, when required, and quality conformance inspection, all electrode types shall be tested using d.c.r.p.

2/ F = flat, V = vertical.

3/ When both tests (3 and 8a) are applicable, radiography of test No. 3 will satisfy the requirements for test No. 8a for class 2 and class 3 electrodes. If test No. 3 is radiographed, test No. 8a may be eliminated for class 2 and class 3 electrodes.

4/ Includes ultimate, yield, elongation, reduction in area.

5/ May be determined from milling sample taken from ruptured ends of tensile specimens or drillings. In event of variances with composition requirements or conflicting results by testing activities, analysis shall be verified by procedures specified in 4.6.8 of MIL-E-22200.

4.5.2 Other tests. Table X summarizes tests, in addition to those specified in table IX required for qualification testing of the electrodes for the QPL and for quality conformance inspection, as applicable.

Table X - Summary of other tests.

Tests	Procedures	Requirements
	MIL-E-22200	
Visual and dimensional	4.6.2.3	3.10 (MIL-E-22200 and 3.5 herein)
Concentricity	4.6.2	3.5.2 (MIL-E-22200)
Dielectric strength[1/]	4.6.3	3.5.4 (MIL-E-22200)
Flaking and cracking of covering[1/]	4.6.4	3.5.5 (MIL-E-22200)
Covering moisture	4.6.9	3.2 (herein)

1/ Required for qualification inspection only.

4.6 Certificate of test. A certificate of quality conformance of the weld tests specified in 4.4.3 (i.e., X-ray, mechanical, and chemical analysis), containing actual quantitative test and inspection results including the range of amperage employed in making the required welds for quality conformance inspection for each accepted lot, heat, and wet mix, as applicable, included in a particular shipment shall be furnished to the consignee with the shipment. The certification shall include test results from the quality

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:23Z
Check the source to verify that this is the current version before use.

conformance test and the wet mix equivalency test as applicable for the shipment. The minimum quality conformance test result data and wet mix equivalency test result data required shall be as shown on figures 3 and 4 and shall be submitted on the forms shown.

4.7 Inspection of preparation for delivery. The packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements.)

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Electrodes shall be packaged in class 1a or class 1c unit containers (50 pounds and under) in accordance with MIL-W-10430, except that when the large containers are specified by the contract or order (see 6.2), 12, 14, and 18-inch electrodes may be supplied in 1b unit containers (50 pounds and under).

5.1.1.1 Special instructions. When electrodes are packaged in class 1a unit containers in accordance with MIL-W-10430, fiberboard or other cushioning material shall be inserted in each end of the unit container to restrict end-wise movement of the electrodes. The cushioning materials shall be dry and in equilibrium with the electrode covering. Tubes in containers shall be of a snug fitting design with the container length not to exceed the rod length by 1/2-inch.

5.1.2 Level C. Packaging shall be sufficient to afford adequate protection against deterioration and physical damage during shipment from the supply source of the using activity. The packaging applied shall insure conformance to the durability requirements specified in 3.5.8 of MIL-E-22200.

5.1.2.1 All electrode types, except as permitted in 5.1.2.2, shall be packaged in class 1 or 2 containers of MIL-W-10430. Unit container weight shall be in accordance with the manufacturer's commercial practice.

5.1.2.2 When specified, types MIL-7015 and MIL-7018 electrodes may be packaged in class 5 unit containers in accordance with MIL-W-10430.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Levels A and B. Electrodes, packaged as specified (see 5.1), shall be packed level A or B as specified in accordance with MIL-W-10430. Palletization, when required (see 6.2), shall be in accordance with MIL-W-10430.

5.2.2 Level C. Electrodes, packaged as specified (see 5.1), shall be packed in containers acceptable to the common carrier and which will insure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers, packing or method of shipment shall comply with Uniform Freight or National Motor Freight Classification Rules or Regulations as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required herein or by the contract or order (see 6.2), interior packages and exterior shipping containers and palletized loads shall be marked in accordance with MIL-W-10430.

5.3.1 Special marking.

5.3.1.1 Interior packages and shipping containers. In addition to the lot identification required by 4.4.6 of MIL-E-22200, identification on interior packages shall include MIL-type, size, specification number, manufacturer's or distributor's name, date of manufacture (month/year), and manufacturer's or distributor's brand or type designation. In addition, all unit packages shall be marked or labeled with the following precautionary information, as a minimum and prominently displayed:

CAUTION
Welding may produce fumes and gases hazardous to health. Avoid breathing these fumes and gases. Use adequate ventilation. See ANSI Z49.1 "Safety in Welding and Cutting" published by the American Welding Society.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:23Z
Check the source to verify that this is the current version before use.

Significant toxic constituents when present in the electrode in greater than trace amounts shall be included in the precautionary information so that normal ventilation can be increased accordingly. These constituents include but are not limited to those itemized in section 8.5 through 8.11 of ANSI Z49.1.

5.3.1.2 Class 1b containers. Each class 1b container on the container end adjacent to the electrode grip and shall be marked "OPEN THIS END".

6. NOTES

6.1 Intended use.

6.1.1 Type MIL-7015 and MIL-7016. These types are intended to cover medium steel electrodes used in welding mild or medium steels such as MIL-S-22698.

6.1.2 Type MIL-8015-C3 and MIL-8016-C3. These types are intended to cover low-alloy electrodes capable of depositing weld metal of higher strength than types MIL-7015 and MIL-7016 and are especially intended for welding grade HT of MIL-S-16113.

6.1.3 Type MIL-10015 and MIL-10016. These types are intended to cover low-alloy electrodes for welding high strength steels such as HY-80 of MIL-S-16216.

6.1.4 All types with suffix RC. These types are intended to cover electrodes for welding pressure and strength welds which will be subjected to irradiation.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class, and size required (see 1.2).
(c) Whether the copper content is restricted as designated by the suffix RC to the MIL-type (see 3.4).
(d) For 3/32 inch or 1/8 inch size electrodes, the length required (see table V).
(e) Whether quality conformance impact tests are required (see table VII, note 1).
(f) Whether stress-relieved quality conformance tests are required (see table VII, note 2).
(g) Levels of packaging and packing required (see 5.1 and 5.2).
(h) When large containers are required (see 5.1.1).
(i) When palletization is required (see 5.2.1).
(j) Special marking required (see 5.3).

6.2.1 Electrodes should be ordered by the pound.

6.3 Inspection after delivery. Post delivery inspection of electrodes to determine conformity to this specification and for acceptance thereof is the responsibility of the consignee.

6.4 THE MARGINS OF THIS SPECIFICATION ARE MARKED "#" TO INDICATE WHERE CHANGES (ADDITIONS, MODIFICATIONS, CORRECTIONS, DELETIONS) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

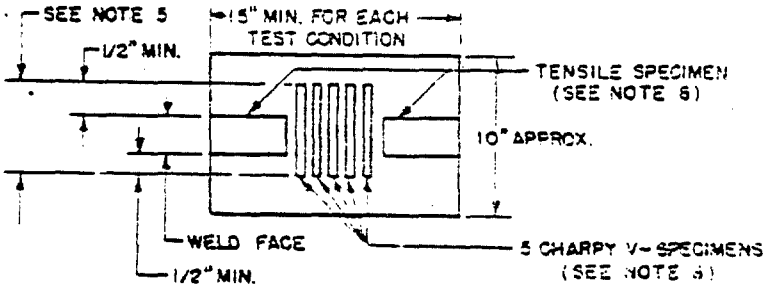
Custodians:
Army - WC
Navy - SH
Air Force - 84

Review activities:
Army - MR, MU, WC
Navy - SH, YD
Air Force - 84

User activities:
Army - AT
Navy - MC, OS

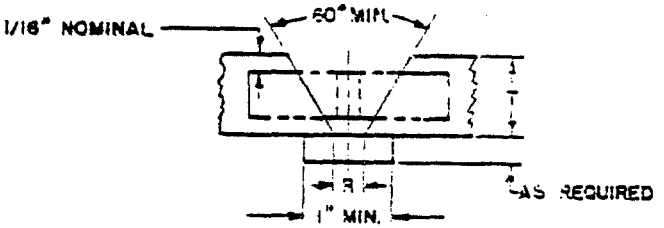
Preparing activity:
Navy - SH
(Project 3439-0317)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:23Z
Check the source to verify that this is the current version before use.



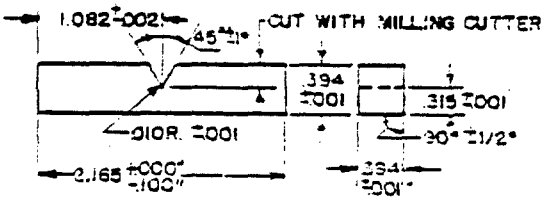
LAYOUT OF TEST PLATE

NOTE: LENGTH MAY BE DECREASED ACCORDINGLY WHERE IMPACTS OR STRESS RELIEVED SAMPLES ARE NOT REQUIRED. WHERE IMPACTS IN THE AS-WELDED AND STRESS-RELIEVED CONDITION BOTH ARE REQUIRED, THE LENGTH MAY BE INCREASED ACCORDINGLY.



GROOVE PREPARATION OF TEST PLATE

SHOWING ORIENTATION OF CHARPY-V TEST SPECIMEN



SH 10104

CUT NOTCH WITH MILLING CUTTER. MAKE FOUR CUTS PER NOTCH IN FOLLOWING ORDER: a) .040; b) .025; c) .010; d) .004 NO IRREGULARITIES IN CONTOUR OF APEX OF NOTCH. NO. 8 POLISH MICRO-INCH FINISH.

Figure 1 - Welded joint for all-weld-metal tension test and impact test (test No.3).

Notes to figure 1:

1. Base metal shall be as specified in 4.5.1.
2. For qualification of the electrodes for the QPL, impact tests in the as-welded and stress-relieved condition are required (see table III).
3. Welds shall be deposited in layers using a 2-1/2 core wire diameter maximum weave build-up sequence. Record and report as specified in table XI. Layers should be approximately 1/8-inch thick, and shall be deposited in such a manner that the final layers include a reinforcement of standard proportions (1/32 inch to 1/8 inch). Welding currents shall be in accordance with the manufacturer's recommendations and the actual values used shall be recorded and reported. Each layer shall be started at the finishing end of the preceding layer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:23Z
Check the source to verify that this is the current version before use.

Notes to figure 1 (con.):

4. After the joint has been tacked, the assembly shall be heated to 225°F ± 25°F prior to welding. After completion of each pass the assembly shall be allowed to cool in still air at room temperature and the subsequent pass shall be started when the temperature of the plate has dropped to 225°F ± 25°F. If it is necessary to interrupt this procedure, the assembly shall be preheated to 225°F ± 25°F.
5. The weld shall be cut as shown resulting in two tensile coupons suitable for 0.505-inch diameter specimens (or 0.252-inch diameter specimens when testing 3/32-inch and 1/8-inch size electrodes) and a center coupon of size sufficient for the required charpy V-notch specimens. NOTE: Impact tests and stress-relieved tests are not required for quality conformance tests unless specified in the procurement document (see tables VII, VIII and IX). Only one tensile specimen and five impact specimens are required when stress-relieving is not required. No base metal shall be removed within 1/2 inch of the edges of the face of the weld by flame cutting. Only sawing or machining shall be used for specimen finishing.
6. One tensile coupon, where required by the procurement document, shall be stress-relieved using the following procedure:

   (a) Employing an electric furnace, the all-weld metal coupon shall be heated to 1150°F ± 25°F, held for 2 hours at temperature and allowed to furnace cool at a rate not to exceed 200°F per hour maximum to a temperature below 500°F.

7. Both the stress-relieved and as-welded coupons shall be machined into 0.505-inch tensile specimens (or 0.252-inch diameter specimens when testing 3/32-inch and 1/8-inch size electrodes). For details see MIL-STD-418 or ASTM A370.
8. The center section of the welded joint shall be used for the preparation of charpy-V-notch impact specimens. The charpy V-notch specimens shall be machined in accordance with details shown on figure 1. Prior to machining specimens, where required by the procurement document, the coupon shall be subjected to the following stress-relieving condition:

   (a) Using an electric furnace, the weld coupon shall be held for 2 hours at 1150°F ± 25°F and then allowed to cool to a temperature below 500°F at a rate not to exceed 200°F per hour.

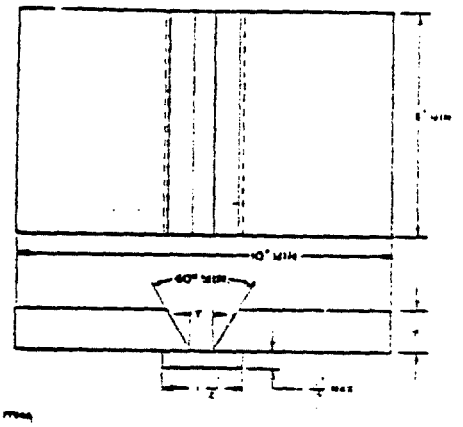
9. Impact properties for the five specimens for each test condition shall be obtained at test temperatures specified in table III herein, plus or minus 3°F. The extreme lowest impact value thus obtained together with the extreme highest value may be disregarded for this test. Two of the three remaining values shall be greater than the specified 20 foot/pound energy level, one of the three may be lower but shall be not less than 15 foot/pounds; the computed average value of these three shall be greater than the specified 20 foot/pound level (see table III herein).

Table XI - Groove weld preparation.

Electrode size	Minimum thickness (T)	Minimum root opening (R)	Number of layers 1/	
			Min.	Max.
(Inch)	(Inches)	(Inch)		
3/32	1/2	1/4	1/	1/
1/8	1/2	1/4	1/	1/
5/32	3/4	1/2	7	9
3/16	3/4	1/2	7	9
7/32	3/4	1/2	7	8
1/4	1	1/2	9	11
5/16	1-1/4	1/2	11	12

1/ Pass and layer sequence shall be reported.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:23Z
Check the source to verify that this is the current version before use.



Electrode size	3/32 (Inch)	1/8 (Inch)	5/32 (Inch)	3/16-5/16 (Inch)
Root opening (Y) 1/	1/4	5/16	3/8	1/2
Thickness (T) 2/	1/4	3/8	1/2	1/2
Weld bead width (max.)	3/8	1/2	5/8	3/4

1/ Tolerance ± 1/16-inch.
2/ Minimum.

Figure 2 - Groove weld usability test (test No. 8a).

Notes to figure 2:

1. Welding shall be accomplished with the test plate in the vertical or flat position as specified in table IX. Welding current shall be in accordance with the manufacturer's recommendations and the actual values shall be recorded and reported. Unless otherwise specified by the procurement document, test shall be done with d.c. only.
2. Base metal shall be in accordance with 4.5.1.
3. The edge preparation shall be flame cut and ground to remove slag and surface irregularities.
4. Welds shall be deposited in approximately 1/8-inch thick layers. Each bead shall contain a start in the area to be evaluated.
5. The reinforcement shall be 3/16-inch maximum.
6. Preheat and interpass temperature shall be to 225° ± 15°F.
7. After completion of the weld, the weld reinforcement and backing strips shall be removed flush with the base plate on both surfaces and the assembly shall be X-rayed in accordance with MIL-STD-271.
8. The radiograph shall meet the X-ray standards of NAVSHIPS 0900-LP-003-9000 as specified in 3.7.1 of MIL-E-22200.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:23Z
Check the source to verify that this is the current version before use.

CERTIFICATION OF QUALITY CONFORMANCE TESTS

Manufacturer or Distributor _____ Customer's Name _____

Address _____

Date _____ Customer's Order No. _____

Specification MIL- _____

Type MIL- _____ Core Wire Heat No. _____

Diameter & Length _____ Lot Identification MIL-E-22200 Para. _____

Inspection Level _____

Lot No. _____ Wet Mix No. _____

Chemical Analysis (Complete)		Mechanical Tests	
	Yield Strength (0.2% offset method)	As-welded	Stress-relieved
Carbon _____	Tensile Strength		
Manganese _____	Elongation (% in __ inches)		
Silicon _____	Reduction of Area (%)		
Phosphorus _____			
Sulfur _____		CHARPY IMPACTS	
Chromium _____		1. _____	1. _____
Nickel _____		2. _____	2. _____
Molybdenum _____		3. _____	3. _____
Vanadium _____		4. _____	4. _____
		5. _____	5. _____

Chemistry was taken from: ☐ Chem Pad ☐ Groove Weld

X-Ray Results _____

Concentricity (%) _____

Covering Moisture _____

Grinding During 3a Test Plate Preparation (see note 3)
Operator Error (Layer Nos.) _____

Groove Weld Test			
Test No.	3	3a	Chem Pad
Amperage	___	_____	_____

We hereby certify that the above material has been tested in accordance with the listed specification and is in conformance with all requirements.

_____

NOTES:

1. Items not applicable to the type electrode involved or exempted by the procurement document shall be marked N/A.
2. This form is applicable to the quality conformance tests required for each lot. Figure 4 forms shall be employed to report and certify the wet mix equivalency tests where applicable for each wet mix of covering mixture in the lot. One copy of the figure 4 form shall be employed for certifying equivalency of each wet mix in a lot where applicable. One copy of the quality conformance certificate (figure 3) for the lot shall be forwarded with copies of the mix equivalency certificates for each additional mix from that lot that is included in the shipment.
3. See paragraphs 3.5.7 and 3.9 of MIL-E-22200.

Figure 3 - Certification of quality conformance tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:23Z
Check the source to verify that this is the current version before use.

CERTIFICATION OF WET MIX EQUIVALENCY TESTS

Manufacturer or Distributor_____ Customer's Name_____

Address_____

Date_____ Customer's Order No._____

Specification MIL-_____

Type MIL-_____

Diameter Size_____ Core Wire Heat No._____

Lot No._____ Wet mix No._____

Sampling

Chemical Analysis

Manganese_____

Silicon_____

Chromium_____

Nickel_____

Molybdenum_____

X-Ray Results_____

Chemistry was taken from:

____ Chem Pad ____ Groove Weld

Test No._____

Chem Pad

Amperage _____ _____ _____

We hereby certify that the above material has been tested in accordance with the latest specification and is in conformance with all requirements.

_____

NOTES:

1. Items not applicable to the type electrode involved or exempted by the procurement document shall be marked N/A.
2. One copy of the figure 4 form shall be employed for certifying equivalency of each wet mix in a lot where applicable. One copy of the quality conformance certificate (figure 3) for the lot shall be forwarded with copies of the mix equivalency certificates for each additional mix from that lot that is included in the shipment.
3. Copper content is required for MIL-types with RC suffix.

Figure 4 - Certification of wet mix equivalency tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:23Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (Mark one)
	☐ VENDOR
b. ADDRESS (Street, City, State, ZIP Code)	☐ USER
	☐ MANUFACTURER
	☐ OTHER (Specify):

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8. DATE OF SUBMISSION (YYMMDD)

DD FORM 32 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS SHEET, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:23Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commander
Naval Ship Engineering Center
Center Building-Sec 61247
Prince George's Center
Hyattsville, MD 20782

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:23Z
Check the source to verify that this is the current version before use.